	CG	CLASS INDEX NO.
	07 ___	

CANDIDATE NAME: ________________________________

TAMPINES JUNIOR COLLEGE
PRELIMINARY EXAMINATION 2008

BIOLOGY 8875/02
Paper 2 Core Paper

THURSDAY 21 AUGUST 2008 **11.30 AM – 1.30 PM (2 hours)**

Additional materials: Answer paper

INSTRUCTIONS TO CANDIDATES

(a) Write your name, Civics group and candidate number in the spaces at the top of this page and on all separate answer paper used.
(b) Write in dark blue or black pen on both sides of the paper.
(c) You may use a soft pencil for any diagrams, graphs or rough working.
(d) Do not use staples, paper clips, highlighters, glue or correction fluid.

SECTION A
Answer **all** questions.
Write the answers in the spaces provided on the question paper.

SECTION B
Answer any **one** question.
Answer on the separate answer paper provided.
The answer should be illustrated by large, clearly labelled diagrams, where appropriate.

At the end of the examination:
Fasten all separate answer paper securely separate from the question paper.

INFORMATION FOR CANDIDATES
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
1	
2	
3	
4	
Section B	
TOTAL	

This question paper consists of 9 printed pages and 1 blank page.

Section A

Answer **all** questions.

QUESTION ONE

Fig. 1.1 represents the molecular structure of tubulin.

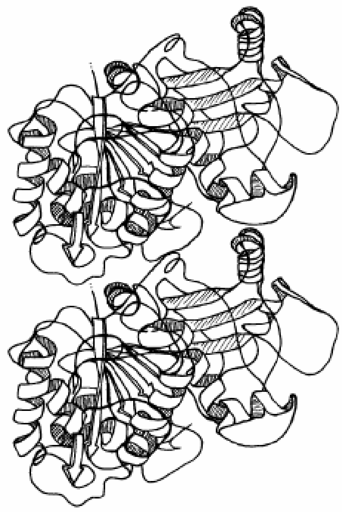


Fig. 1.1

(a) Describe the secondary, tertiary and quaternary structure of tubulin shown in Fig. 1.1.

____

____

____ [3]

(b) Tubulin is involved in the construction of microtubules.

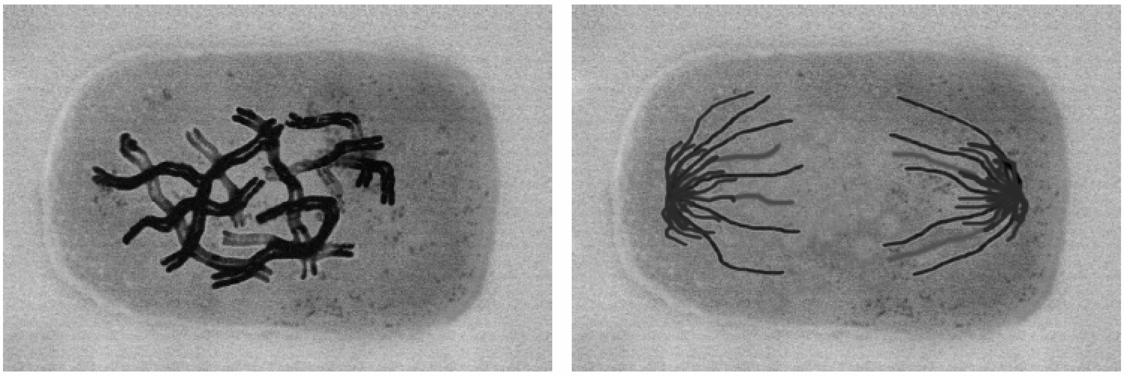
(i) Where in the cell do microtubules originate?

____ [1]

(ii) Describe the role of microtubules in M phase of the cell cycle.

____ [1]

Fig. 1.2 shows two stages of mitosis in a cell from a root tip of *Allium cepa.*



D E

Fig. 1.2

(c) Describe what happens to the chromosomes during mitosis between the stage shown in **D** and the stage shown in **E**.

[3]

(d) Explain how the events in the mitotic cell cycle ensure that all the cells in the root are genetically identical.

[3]

[Total: 11]

QUESTION TWO

Fig. 2 shows events occurring during the synthesis of a protein that is secreted from a cell.

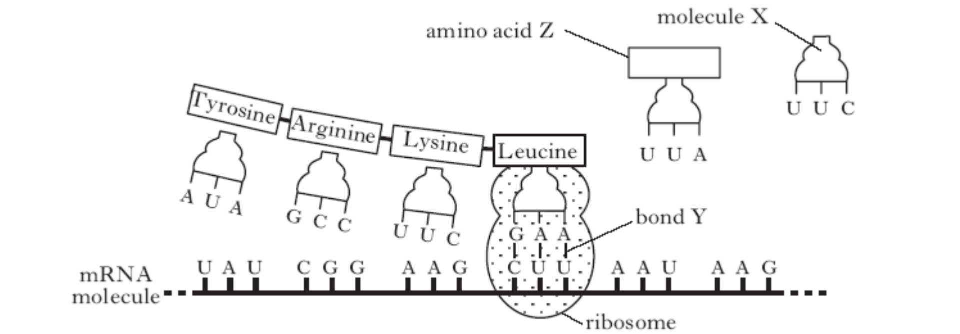


Fig. 2

(a) (i) Name molecule **X**. ______________________________________ [1]

(ii) Name bond **Y**. ____ [1]

(b) What name is given to a group of three bases on mRNA that codes for an amino acid?

____ [1]

(c) Give the sequence of DNA bases that codes for amino acid Z.

____ [1]

(d) Describe the roles of the endoplasmic reticulum and the Golgi apparatus between the synthesis of the protein and its release from the cell.

Endoplasmic reticulum ____

____ [1]

Golgi apparatus ____

____ [1]

(e) The table contains some information about the structure and function of proteins.
Add information to the boxes to complete the table.

Protein	Structure (Globular or Fibrous)	Function
Cellulase		
Collagen		Structural protein in skin

[3]
[Total: 9]

QUESTION THREE

The following figure shows the main stages in the polymerase chain reaction (PCR).

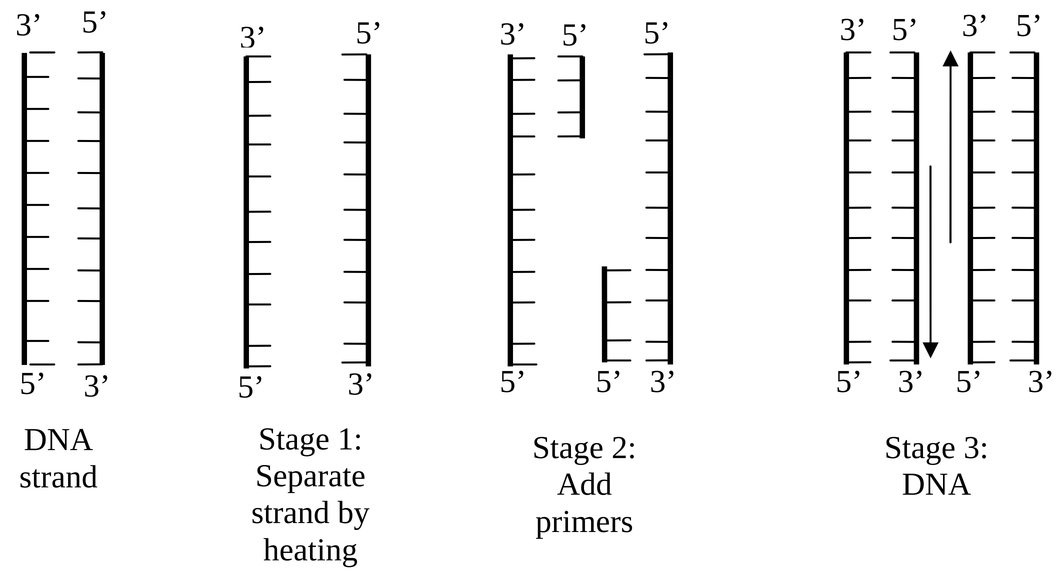


Fig. 3

(a) With reference to **Fig. 3**, explain what is happening at each of the three stages.

(i) **Stage 1:**

____

____

____

____ [2]

(ii) **Stage 2:**

____

____

____

____ [2]

(iii) **Stage 3:**

[2]

(b) Suggest why **one advantage** and **one disadvantage** of using **PCR**.

[2]

(c) Explain the **advantages** of using **Taq DNA polymerase**.

[2]

[Total: 10]

QUESTION FOUR

Sickle cell anaemia is a condition where the haemoglobin in the red blood cells can no longer carry oxygen efficiently at low oxygen concentrations. Heterozygous individuals show some symptoms of anaemia and appear to be at a potential disadvantage, but they have some resistance to malaria.

In **Fig. 4**, **A** shows the distribution of the malarial parasite in parts of East Africa below 1500 metres, whilst **B** shows the area in which more than 15% of the adults in the population are heterozygous for the sickle cell condition.

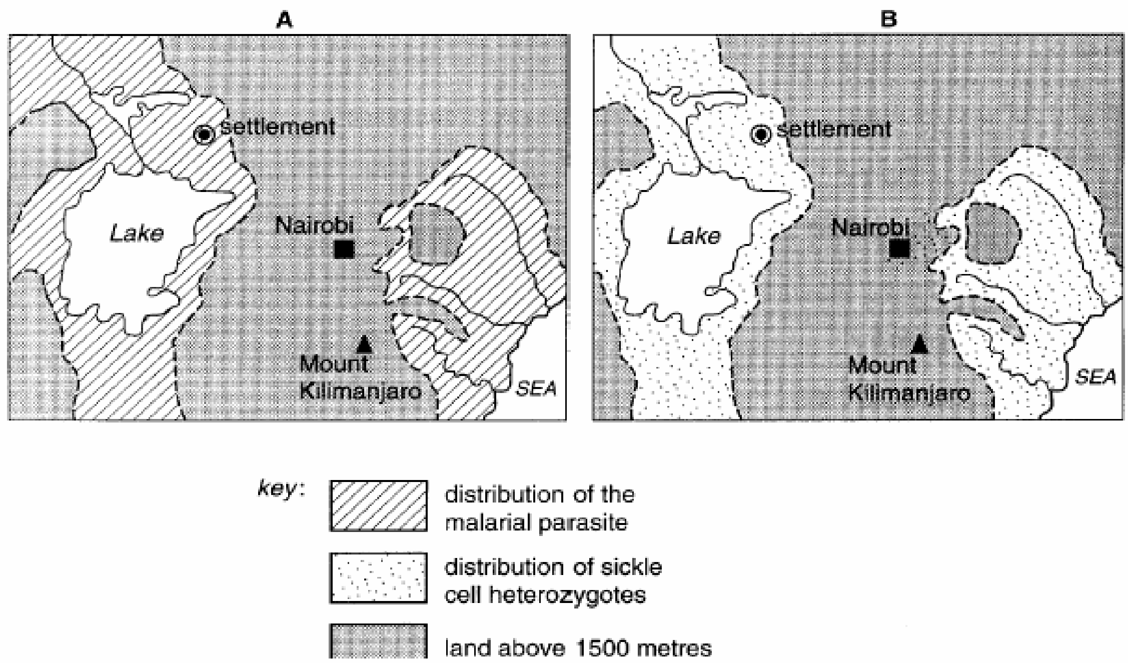


Fig. 4

(a) Suggest why, and under what conditions, the red blood cells become sickle-shaped.

____

____

____

____ [2]

(b) The individual who is heterozygous for the sickle cell allele has the genotype $\mathbf{Hb^A\ Hb^S}$.

Using the symbols given, illustrate by means of a genetic diagram the probability of inheriting sickle cell anaemia from parents who are heterozygous for the sickle cell allele.

Probability: ____ [1]

Derivation:

[3]

(c) With reference to **Fig. 4**,

Describe, with reasons, the distribution of the sickle cell heterozygotes.

____

____

____

____ [2]

(d) If the inhabitants of the settlement indicated on **Fig. 4** were moved to an area where malaria did not exist, suggest what might happen to the population of heterozygotes in future generations. Explain your answer.

____

____

____

____ [2]

[Total: 10]

Section B

Answer **one** question.

Write your answers to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a), (b)** etc., as indicated in the question.

<u>QUESTION FIVE</u>

(a) Describe features of a plasmid that made it suitable as a vector. [4]

(b) Using a named example, outline the process of cloning a eukaryotic gene into *E.coli* cells to produce eukaryotic protein in genetic engineering. [10]

(c) Discuss the goals and implications of the human genome project, including the benefits and difficult ethical concerns. [6]

<u>QUESTION SIX</u>

(a) Explain how ATP is synthesised at the mitochondrial inner membrane from one molecule of glucose for cellular activities, with reference to how the inner membrane is adapted for the synthesis. [12]

(b) Compare and contrast the Krebs cycle and the Calvin cycle. [8]